

MySQL Data Types Reference

Comprehensive Quick Guide

Numeric Types

Data Type	Description / Range
BIT(M)	A bit-field type. M indicates the number of bits per value, from 1 to 64.
TINYINT	A very small integer. Signed range: -128 to 127. Unsigned: 0 to 255.
BOOL / BOOLEAN	Synonyms for TINYINT(1). Zero is false, non-zero is true.
SMALLINT	A small integer. Signed range: -32768 to 32767. Unsigned: 0 to 65535.
MEDIUMINT	A medium-sized integer. Signed: -8,388,608 to 8,388,607. Unsigned: 0 to 16,777,215.
INT / INTEGER	A normal-size integer. Signed: -2,147,483,648 to 2,147,483,647.
BIGINT	A large integer. Signed: -9,223,372,036,854,775,808 to 9,223,372,036,854,775,807.
DECIMAL / NUMERIC	Fixed-point number. Precision (total digits) and Scale (decimal places).
FLOAT	A small floating-point number (4 bytes).
DOUBLE	A normal-size floating-point number (8 bytes).

String (Character) Types

Data Type	Description / Range
CHAR(M)	A fixed-length string (0 to 255 characters). Always right-padded with spaces.
VARCHAR(M)	A variable-length string (0 to 65,535 characters).
TINYTEXT	A very small non-binary string (max 255 characters).
TEXT	A small non-binary string (max 65,535 characters).
MEDIUMTEXT	A medium-size non-binary string (max 16,777,215 characters).
LONGTEXT	A large non-binary string (max 4,294,967,295 characters).
BINARY(M)	Similar to CHAR, but stores binary byte strings.
VARBINARY(M)	Similar to VARCHAR, but stores binary byte strings.
TINYBLOB	A very small BLOB (Binary Large Object).
BLOB	A small BLOB.
MEDIUMBLOB	A medium-size BLOB.
LOB	A large BLOB.
ENUM	An enumeration. A string object that can have only one value from a list.

Data Type	Description / Range
SET	A set. A string object that can have zero or more values from a list.

Date and Time Types

Data Type	Description / Range
DATE	A date. Format: YYYY-MM-DD. Range: '1000-01-01' to '9999-12-31'.
DATETIME	A date and time combination. Format: YYYY-MM-DD hh:mm:ss.
TIMESTAMP	A timestamp. Range: '1970-01-01 00:00:01' UTC to '2038-01-19 03:14:07' UTC.
TIME	A time. Format: hh:mm:ss.
YEAR	A year in 4-digit format. Range: 1901 to 2155.

Spatial Data Types

Data Type	Description / Range
GEOMETRY	A type that can store geometry values of any type.
POINT	A point in 2D space.
LINestring	A curve with linear interpolation between points.
POLYGON	A planar polygon.
JSON	

Data Type	Description / Range
	Stores JSON data. Provides automatic validation and optimized storage.